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RESEARCH ARTICLE

Agentic AI Arrives: How Gen Z Adopts Autonomous AI Agents

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ABSTRACT In the era of agentic artificial intelligence (AI), autonomous systems capable of operating in the background and intelligently performing tasks on behalf of users are rapidly emerging. However, little is known about the factors influencing user adoption of agentic systems. This study analyzes adoption intentions toward AI agents among members of Generation Z (born in 1997 or later), who are often described as early adopters of new technologies. The analysis is based on data collected through an online survey in Germany with 243 participants. Despite being the generation most accustomed to digital technologies, only about half of the respondents (49.4%) have previously interacted with agentic AI. Nevertheless, nearly 39.6% expressed interest in engaging with such technology, while only 11.1% would not want to use agentic AI at all. The PLS-SEM analysis revealed that perceived usefulness was the strongest predictor of intention to use, while perceived risk had both a significant negative direct effect on intention and an indirect effect via trust. A key finding is that although respondents expressed concerns about data privacy and reliability, (the lack of) trust did not significantly influence their intention to use AI agents. The NCA results further suggest that perceived usefulness and perceived humanness are necessary conditions for user adoption. These insights contribute to a deeper understanding of Gen Z's attitudes toward AI agents and can inform strategies to accelerate the uptake as this technology continues to mature. We also discuss managerial and policy recommendations to guide responsible, transparent, and trustworthy application of agentic AI in society.

INDEX TERMS Agentic AI, AI super assistants (AISA), human–AI interaction, AI adoption, Gen Z.

I. INTRODUCTION

Artificial intelligence (AI) has become deeply embedded in everyday life, supporting tasks such as information retrieval, content curation, and decision-making across personal, professional, and institutional contexts [1]. Yet recent technological advances mark a profound shift: from reactive, task-specific systems to agentic AI, defined as autonomous, goal-oriented agents capable of planning, acting, and learning with minimal human input [2]. Compared to established conversational AI chatbots, agentic systems like ChatGPT (agent mode), Claude (Chrome extension), and Manus act autonomously to complete multi-step processes rather than merely responding to prompts. As a result, they enable users to accomplish more complex tasks with less direct effort [3]. Despite its growing relevance and real-world impact, empirical research on whether and why consumers would adopt agentic AI systems remains scarce. To the best of our knowledge, only one study has so far examined acceptance of highly autonomous AI agents – termed AI super assistants (AISA) – in the Philippines, identifying both drivers and barriers of adoption [4]. The findings suggest that users do not perceive highly autonomous AI agents as overwhelming but rather feel capable of engaging with their new functionalities and comfortable with increased levels of agency. Moreover, acceptance appears to be driven by positive experiential and performance-related expectations. However, the study only assessed stated preferences at a time when AI agents were

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not yet available, limiting insights into real user behavior. The present study offers a novel perspective by investigating the adoption of autonomous AI agents among Generation Z (Gen Z), a cohort typically considered early adopters of emerging digital technologies. As AI-based tools and services are rapidly diffusing into everyday applications, Gen Z represents a relevant group because their technology habits, expectations, and risk perceptions are likely to shape future adoption patterns and market developments [5]. We select Germany as a case because it combines a high level of digital maturity with strict data protection standards and an active public and policy discourse on the integration of digital technologies into society [6]. Using survey data from 243 respondents in Germany and combining partial least squares structural equation modeling (PLS-SEM) with necessary condition analysis (NCA), the present research uncovers the latent factors that shape interest, intention, and actual use of AI agents, including insights from respondents who have already used agentic AI systems in practice.

II. LITERATURE REVIEW AND HYPOTHESES DEVELOPMENT

Research on AI adoption suggests that traditional acceptance models have limitations in explaining user responses to complex and autonomous AI systems [7], [8]. We therefore adopt a theory-informed framework based on well-established constructs tailored to the characteristics of agentic AI.

Perceived humanness (PH) is expected to play a central role in how users evaluate and interact with AI agents. Research demonstrates that human-like communication cues foster relational perceptions and increase trust in AI-enabled interactions by making systems appear more socially responsive and relatable [9]. Humanlike behaviors also enhance users' sense of social presence, authenticity, and emotional connection with AI systems, which shapes how credible and intuitively supportive these agents are perceived to be [10]. As agentic AI increasingly engages in autonomous and context-sensitive interactions, perceived humanness becomes a key factor in reducing psychological distance and facilitating positive user evaluations. Accordingly, we propose the following hypotheses.

H1a: Perceived humanness positively affects trust.

H1b: Perceived humanness positively affects perceived usefulness.

In contrast, perceived risk (PR) is a central barrier to the adoption of AI systems. Prior studies show that concerns about uncertainty, data misuse, and the lack of algorithmic transparency reduce users' confidence in AI-based services and hinder engagement. In highly personalized or autonomous settings, perceived risk becomes even more salient, as users fear potential misuse of sensitive information or unintended system behavior [4], [11]. Research on voice-controlled AI similarly demonstrates that privacy concerns and perceived vulnerability undermine trust and discourage use [12]. Taken together, these findings suggest that PR erodes users' sense of safety and reliability in AI interactions. Therefore, we propose the following hypotheses.

H2a: Perceived risk negatively affects trust.

H2b: Perceived risk negatively affects intention to use AI agents.

Perceived usefulness (PU) captures the extent to which users believe that AI agents help them accomplish tasks more effectively, efficiently, and with less cognitive effort. Previous research identifies PU as a key driver of technology adoption intentions, particularly among younger adults and students, who tend to prioritize efficiency gains and practical value when evaluating digital tools. Recent evidence from university settings further shows that higher PU strongly increases behavioral intention to use ChatGPT [13], indicating that users are more willing to adopt AI systems when clear performance benefits are evident. Given the autonomous and task-oriented nature of agentic AI, similar effects can be expected in this context. Therefore, we hypothesize the following.

H3: Perceived usefulness positively affects intention to use AI agents.

Trust (TR) is another central construct in the acceptance of AI technologies, particularly when systems operate autonomously and make decisions on behalf of users. Prior research suggests that higher levels of TR increase users' willingness to adopt AI systems [14], promote greater reliance on and engagement with generative AI systems [15], reinforce confidence in autonomous and decision-making technologies [16], and strengthen adoption intentions of AI agents [17]. Trust reduces perceptions of uncertainty, vulnerability, and loss of control by signaling that the AI system will act reliably and in the user's best interest. In the context of agentic AI, where users delegate tasks and decisions to autonomous agents, trust becomes especially relevant for shaping positive evaluations and behavioral intentions. Based on these insights, we expect the following relationship.

H4: Trust positively affects intention to use AI agents.

Taken together, the proposed theory-informed research model and the corresponding hypotheses are presented in Fig. 1.

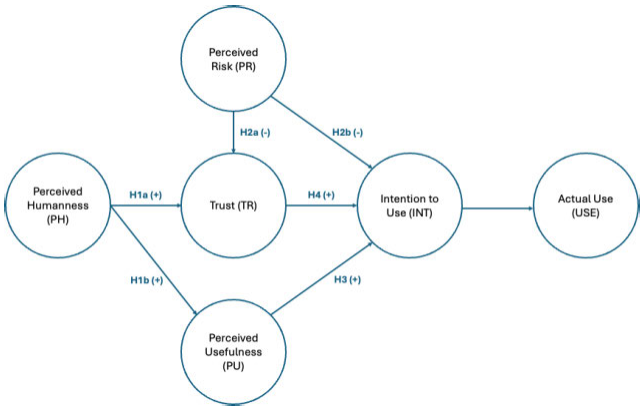

FIGURE 1. The proposed research model.

TABLE 1. Measurement items.

Code	Item description	Supporting literature
PH1	AI's responses feel natural	[4][9]
PH2	AI has a humanlike response	
PH3	AI's responses do not feel machine-like	
PH4	AI reacts in a very human way	
PR1	It is risky to provide personal information to an AI agent	[4]
PR2	There will be much uncertainty associated with providing personal information to an AI agent	
PR3	There will be much potential loss associated with providing personal information to an AI agent	
PR4	I have concerns about the reliability and trustworthiness of the services provided by an AI agent	
TR1	I trust that an AI agent can offer information and service that's best of my interest	[14]
TR2	I trust that my personal data is protected from potential abuse when using an AI agent	
TR3	I trust that my privacy is protected when using an AI agent	
TR4	I trust that authorities exert effective control over organizations and companies providing AI agent services	
PU1	Using an AI agent would enable me to accomplish tasks more quickly	[14]
PU2	Using an AI agent would improve my performance at accomplishing tasks	
PU3	Using an AI agent for accomplishing tasks would increase my productivity	
PU4	I find an AI agent useful for me to accomplish tasks	
INT1	I am willing to receive an AI agent	[4]
INT2	I would feel happy to interact with an AI agent	
INT3	I am likely to interact with an AI agent	

III. DATA AND METHODS

A. SURVEY DESIGN AND MEASUREMENT

The questionnaire was developed in English and translated into German by the first author, a native speaker. Informed consent was obtained from all participants. The instrument first informed about the purpose and structure of the online survey as well as data protection policies. It was explained that participation was voluntary, responses were anonymous and confidential, and withdrawal from the study was possible at any time. After passing the screening questions, eligible participants received a detailed description of AI agents. This included their key features, typical applications, and concrete examples (i.e., ChatGPT's agent mode). To better support participants' understanding of the concept, the information was also accompanied by an explanatory infographic. At the end of the survey, participants were given the opportunity to provide additional feedback on the topic through an open-ended question.

Sociodemographic items were aligned with the response categories used in the national census in Germany. Actual use was assessed with the following item: "Have you already used an AI agent?" and the corresponding response options:

1 = No, and I am not interested (n=30)
2 = No, but I am interested in trying it in the future (n=110)
3 = Yes, I have tried an AI agent, but I am not currently using one (n=32)
4 = Yes, I occasionally use an AI agent (n=46)
5 = Yes, I regularly use an AI agent (n=56)

The theory-driven latent constructs used validated items from prior studies in the technology adoption and human–AI interaction literature [4], [9], [14] and were measured on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree) (Table 1).

TABLE 2. Sample characteristics (N=243).

Variable	Category	Mean (SD)	% (n)
Gender	Female	-	50.6 (123)
	Male	-	49.4 (120)
	Non-binary	-	0 (0)
Age	[18-28]	24.6 (2.32)	-
Household net income	Less than 1,500€	-	72.0 (175)
	1,500€ to less than 2,600€	-	15.2 (37)
	2,600 to less than 5,000€	-	11.5 (28)
	5,000€ and more	-	1.2 (3)
Place of residence	Rural Area	-	18.1 (44)
	Small Town	-	12.3 (30)
	Medium-Sized Town	-	13.6 (33)
	Large City	-	30.9 (75)
	Metropolis	-	25.1 (61)

B. PARTICIPANTS

Respondents were recruited in November 2025 via Survey-Circle, an online platform commonly used for peer-based survey recruitment. Registered users were able to access the link to the online survey if they met the predefined target group criteria. After removing "speeders", "straightliners", and participants that did not pass the attention checks, a total of 243 valid responses were collected (Table 2). The average completion time was 4:16 minutes.

Due to applied quotas, the sample was balanced with respect to gender (50.6% female, 49.4% male). Participants were young adults of Gen Z (M_age = 24.6, SD = 2.32) with predominantly low to moderate household incomes. Most respondents lived in urban environments, with large

cities and metropolitan areas representing over half of the sample.

C. DATA ANALYSIS

To analyze the collected primary data, we applied a combined partial least squares structural equation modeling (PLS-SEM) and necessary condition analysis (NCA) approach. This methodological combination has been used in recent technology adoption studies to jointly examine sufficiency- and necessity-based relationships [e.g., [4], [11]]. PLS-SEM was employed to evaluate the measurement model and to test the hypothesized structural relationships [18]. NCA complemented these results by identifying whether specific constructs represent necessary conditions and by determining bottleneck levels required for specific levels of adoption to occur [19]. Together, this approach provides a robust validation of both the measurement properties and the underlying causal structure of the proposed model.

We follow established guidelines to adequately integrate the two methods [20]. The procedure starts with PLS-SEM, as NCA builds on the latent variable scores obtained from the PLS-SEM analysis. Note that our sample size is sufficient to ensure adequate statistical power for both methods [20], and that previous studies report comparable sample sizes [e.g., [21], [22]]. Data analysis was carried out in R (v4.5.1) using the *seminr* and *NCA* packages.

IV. RESULTS

A. MEASUREMENT MODEL

The measurement model assessment evaluated the reliability, convergent validity, and discriminant validity of all latent constructs.

The factor loadings of all items are above the commonly accepted threshold of 0.60 for exploratory research, ranging from 0.617 to 0.942. This demonstrates that each indicator loads well on its intended construct.

For discriminant validity assessment, the Heterotrait–Monotrait ratio (HTMT) was used instead of the Fornell–Larcker criterion, as it is considered a more stringent and reliable method in variance-based SEM [23]. The HTMT values range from 0.159 to 0.673, well below the recommended threshold of 0.85 (Table 3).

All Cronbach's alpha and composite reliability (CR) values exceed the recommended cut-off of 0.70, while the Average Variance Extracted (AVE) values are above 0.50 (Table 4). This indicates satisfactory reliability and convergent validity.

Moreover, the inner VIF values are all well below the recommend benchmark of 5 (PR = 1.205, TR = 1.223, PU = 1.057), suggesting that multicollinearity is not a concern.

TABLE 3. Discriminant validity assessment.

	PH	PR	TR	PU	INT
PH					
PR	0.159				
TR	0.207	0.461			
PU	0.501	0.189	0.242		
INT	0.469	0.349	0.263	0.673	

TABLE 4. Reliability and convergent validity assessment.

Construct	Item	Loading	α	CR	AVE
Perceived humanness	PH1	0.800	0.737	0.835	0.562
	PH2	0.791			
	PH3	0.776			
	PH4	0.617			
Perceived risk	PR1	0.840	0.819	0.880	0.649
	PR2	0.883			
	PR3	0.747			
	PR4	0.743			
Trust	TR1	0.698	0.825	0.884	0.660
	TR2	0.894			
	TR3	0.908			
	TR4	0.728			
Perceived usefulness	PU1	0.855	0.904	0.933	0.777
	PU2	0.900			
	PU3	0.899			
	PU4	0.872			
Intention to use	INT1	0.942	0.877	0.925	0.805
	INT2	0.816			
	INT3	0.927			

B. STRUCTURAL MODEL

After confirming the reliability and validity of the measurement model, we proceeded to evaluate the structural model and the hypothesized relationships. Figure 2 illustrates the results of the structural model. PH had a small but statistically significant positive effect on TR ($\beta = 0.128$, $p < .05$) and a strong positive effect on PU ($\beta = 0.413$, $p < .001$). In addition, the total effect of PH on INT was significant ($\beta = 0.235$, $p < .001$), indicating an indirect influence via PU.

PR showed a substantial negative effect on TR ($\beta = -0.389$, $p < .001$) and also directly reduced INT ($\beta = -0.194$, $p < .001$). In contrast, TR did not significantly predict INT ($\beta = 0.030$, $p > .05$). PU emerged as the strongest predictor of INT ($\beta = 0.560$, $p < .001$), which in turn had a strong positive effect on USE ($\beta = 0.516$, $p < .001$).

Accordingly, Hypotheses 1 to 3 were supported by the results, whereas Hypothesis 4 was not supported. During model estimation, we also controlled for socioeconomic variables, including age, gender, income, and place of residence. However, none of these variables showed a significant effect on intention to use.

Overall, the model explained 18% of the variance in TR, 17% in PU, 40% in INT, and 27% in USE. In line with established methodological guidance, these values indicate an acceptable to satisfactory level of predictive power for a behavioral adoption model, particularly in consumer and technology usage contexts [18]. Since perceptual constructs and actual use behavior for emerging technologies are influenced by additional contextual and situational factors, lower explained variance is generally expected and still regarded as meaningful.

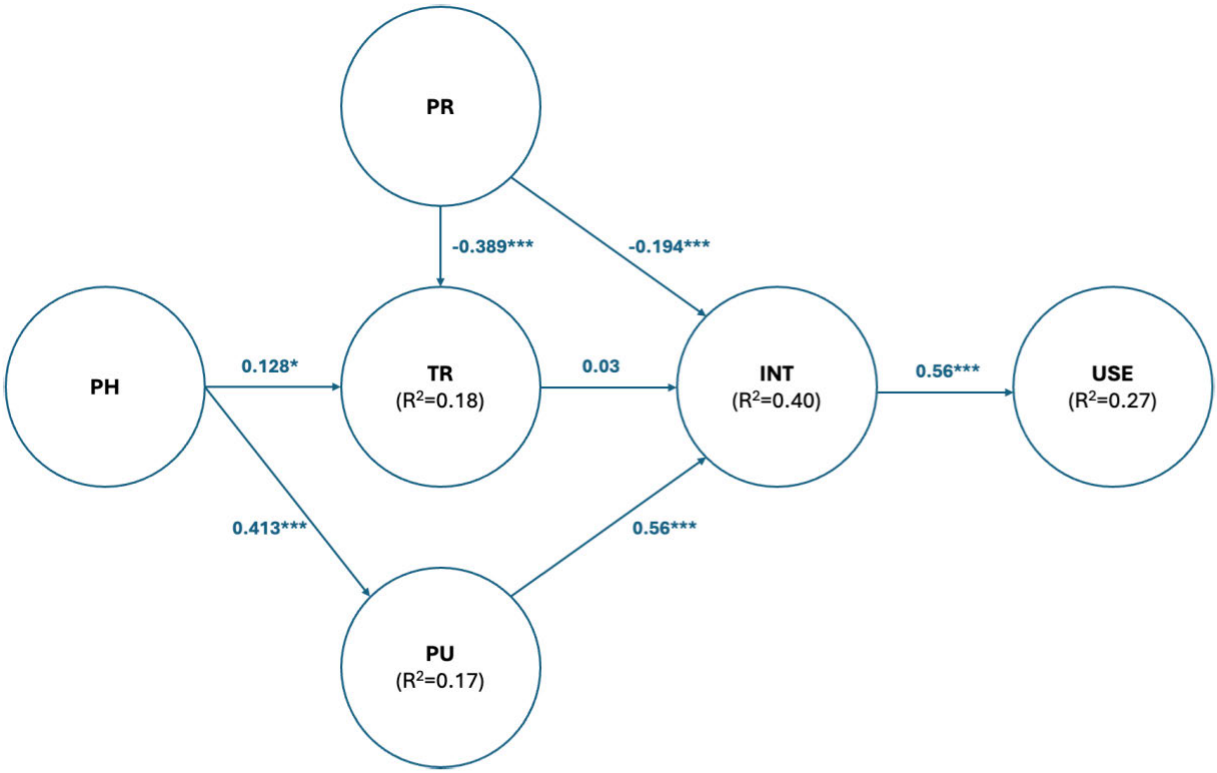

FIGURE 2. PLS-SEM results.

C. NECESSARY CONDITION ANALYSIS

The NCA was applied only to constructs that could theoretically function as necessary conditions for INT, namely PH, TR, and PU. PR was not considered in the necessity logic because higher levels of PR reduce – rather than enable – INT, making it conceptually incompatible with the notion of a necessary condition.

In line with prior recommendations, the ceiling regression with free disposal hull (CR-FDH) technique was used, as it represents the default approach for discrete data with multiple measurement levels [19]. In the scatter plots (Fig. 3), the CR-FDH ceiling line is illustrated in yellow. It smooths the stepwise ceiling envelopment with free disposal hull (CE-FDH) ceiling line, shown in red.

The results show that PH and PU operate as necessary conditions for INT to occur, illustrated by the upper-left empty region in the scatter plots that represents infeasible combinations of low PU/PH and high INT (Fig. 3a+b). PU exhibits a medium necessary effect (CR-FDH = 0.284, $p < .001$), while PH shows a smaller but significant necessary effect (CR-FDH = 0.152, $p < .001$). In contrast, trust TR does not demonstrate a significant necessary effect (CR-FDH = 0.000, $p > .05$) (Fig. 3c).

We further perform a bottleneck analysis, which identifies the minimum levels of necessary conditions that must be met for different levels of INT to occur. The results illustrate that increasing levels of INT require proportionally higher minimum levels of PH and PU (Table 5). This indicates that both constructs function as enabling conditions that must reach certain thresholds before higher levels of intention can occur. For example, achieving an INT level of 60 requires minimum levels of 18.7 for PH and 32.6 for PU. As INT levels increase, these minimum requirements rise accordingly, highlighting that insufficient levels of PH or PU constrain adoption regardless of other favorable conditions. In contrast, TR does not impose such bottlenecks, confirming that it is not a necessary condition for INT.

TABLE 5. Bottleneck analysis.

INT	PH	TR	PU
0	NN	NN	7.6
10	1.0	NN	11.8
20	4.5	NN	15.9
30	8.1	NN	20.1
40	11.6	NN	24.3
50	15.1	NN	28.4
60	18.7	NN	32.6
70	22.2	NN	36.8
80	25.7	NN	40.9
90	29.2	NN	45.1
100	32.8	NN	49.3

V. DISCUSSION AND CONCLUSION

As one of the first empirical investigations into the drivers of AI agent adoption, this study contributes to the emerging literature on agentic AI. It confirms previous research on Gen Z's high interest in AI technologies [5], [24] and highlights how this cohort evaluates and engages with a novel generation of autonomous AI agents.

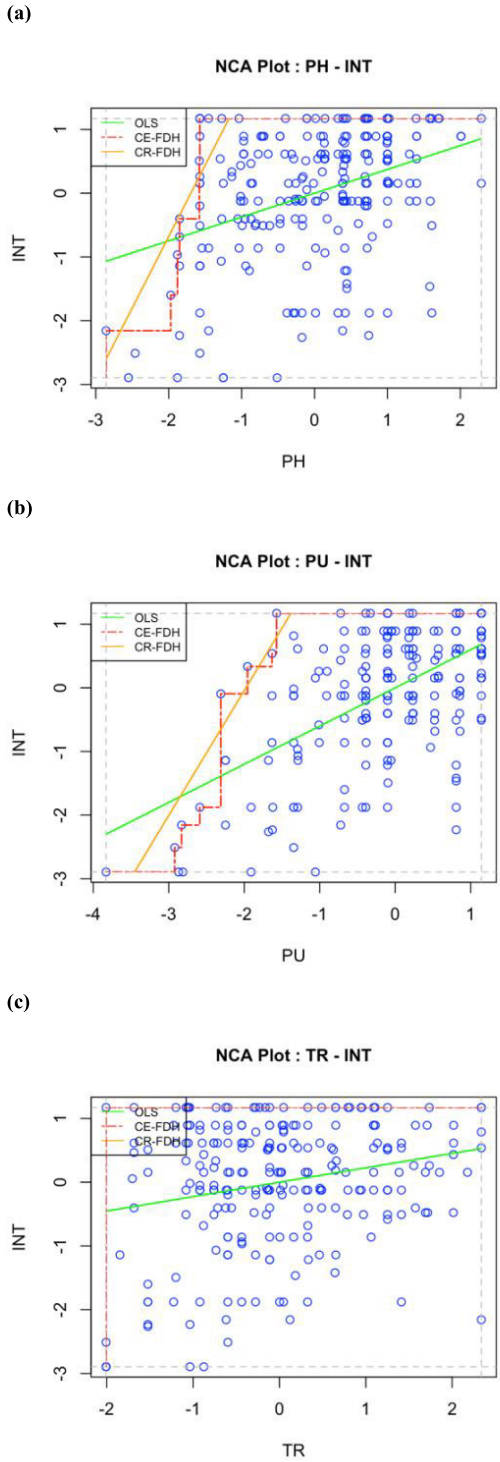

FIGURE 3. NCA scatter plots.

The results show that both perceived usefulness and perceived humanness function as sufficiency-based predictors of adoption and also operate as necessary conditions for higher levels of intention to use. Hence, while Gen Z evaluates autonomous AI systems primarily through a pragmatic lens, they also expect AI agents to communicate and behave in ways that feel intuitive and socially aligned.

In contrast, perceived risk reduces trust and intention, showing that concerns about data practices, reliability, and potential misuse still play an important role. This supports earlier findings that higher risk perceptions weaken users' sense of safety and credibility in AI interactions [12]. However, for Gen Z, such concerns appear to be balanced against and often outweighed by the strong usefulness they perceive/expect in AI agents.

Indeed, although respondents expressed concerns about data privacy and reliability, trust did not significantly influence intention, suggesting that Gen Z users may prioritize the perceived utility of AI agents over potential concerns.

A. PRACTICAL IMPLICATIONS

The findings of this study offer several practical implications for AI providers, marketers, and policymakers who aim to support the responsible and successful introduction of agentic AI into society, particularly in the context of young users.

For companies developing agentic AI, the results highlight the importance of designing systems that deliver clear and immediate functional value. It is imperative to emphasize features that save time, automate routine tasks, and simplify complex processes. Demonstrating tangible benefits through interactive tutorials and real-world use cases can strengthen perceived usefulness among young adults.

Providers should also prioritize human-centered interaction design. For example, systems should incorporate adaptive communication styles, contextual responsiveness, and emotionally aware feedback. Enhancing these aspects may help young adults form a sense of connection and comfort with AI agents, thereby increasing sustained engagement.

Given that Gen Z often seeks tools that enhance productivity and reduce friction in everyday life [24], communication strategies should focus on scenarios where AI agents support learning, work, or personal organization. While these insights are derived from a German sample, similar expectations may apply to young users in comparable digital and regulatory contexts. At the same time, transparent communication about limitations, error cases, and data practices is crucial for building credibility, even if trust is not the key driver of initial adoption.

Organizations integrating agentic AI into their services should consider that young users may adopt such systems despite unresolved concerns. This creates both opportunities and responsibilities. On one hand, managers can leverage the openness of Gen Z by integrating autonomous agents into customer service, education, or productivity tools. On the other hand, they must ensure that convenience does not overshadow the need for robust risk mitigation, user control, and transparent design.

From a broader societal perspective, the finding that trust did not significantly shape intention stands out as especially important. It suggests that the responsibility for ensuring responsible agentic AI use cannot be left to individual users alone. Instead, it must be addressed through regulation. Policymakers must prioritize clear guidelines for

transparency, data governance, and accountability in agentic AI systems [25], particularly in regulatory environments such as Germany that place strong emphasis on data protection and user rights. This includes standardized disclosure of autonomy levels, audit mechanisms for data use, and strong regulatory oversight to prevent misuse and protect vulnerable users from potential harms. Moving forward, ensuring responsible, transparent, and trustworthy deployment of agentic AI must be a key priority as these systems become increasingly integrated into everyday life.

B. LIMITATIONS AND FUTURE RESEARCH

This study is not without limitations. First, all measures relied on self-reported data, which may not fully capture real-world behavior. Second, because participation was voluntary, the data may be subject to non-response bias, as individuals with stronger opinions or higher interest in AI technologies may have been more likely to complete the survey.

In future research, scholars can address these limitations by collecting actual usage data and employing longitudinal approaches.

Conceptually, future research could extend the present study by differentiating between distinct types and application domains of AI agents, such as agents used for sales, research, education, or personal productivity.

Methodologically, fuzzy-set qualitative comparative analysis (fsQCA) can be used to identify multiple configurational pathways through which different combinations of conditions jointly lead to high adoption intentions or actual use of agentic AI. This approach is frequently combined with PLS-SEM and NCA to provide a more holistic understanding of complex and asymmetric causal relationships [26].

Moreover, qualitative approaches can provide richer insights. In the open-ended survey question, strong perceived value and interest were highlighted. As one respondent (f, 22) expressed: "AI agents are a great help in my everyday life, and I wouldn't want to go without them anymore."

More research is also needed on non-adoption factors. One respondent (m, 26) emphasized concerns regarding reliability: "AI models are increasingly hallucinating. They also provide false information with complete confidence. There is no transparency about how the data is processed or who has access to it. I mostly use them as a new kind of Google (searching through their sources for the correct information) or for programming. But I definitely do not trust them."

Another respondent (m, 28) focused on ethical and infrastructural concerns: "The technology is exciting and has great potential, for example for optimizing part-time learning or coordinating household tasks (smart home, cleaning schedule, etc.). However, the dominance of U.S.-based services is concerning, as they are not known for data minimization. A European-developed, supervised, and hosted alternative would immediately motivate me to use it."

These examples confirm that, beyond utility, factors such as data governance, transparency, and ethical considerations play a crucial role in shaping acceptance. Building on the above, scholars can employ in-depth qualitative methods to further explore the underlying motivations, concerns, and expectations associated with AI agents.

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